

How Does CEO Age Affect Firm Risk?

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Abstract

Previous research demonstrates that older CEOs are associated with lower firm equity risk and fewer R&D expenditures. We examine the risk-taking behavior of CEOs and find that not only do older CEOs invest in less R&D, they do so sub-optimally, in the sense that CEO age distorts the effect of q on R&D investment decisions. We find that it is specifically through the channel of R&D investment that CEO age is associated with the reduction in firm equity risk. Consistent with the sub-optimality of these distortions, firms with more effective corporate governance do not exhibit these associations.

Keywords CEO age; Corporate governance; Idiosyncratic risk; R&D expenditure; Systematic risk

JEL Classification: G30, G32

1. Introduction

Several studies tie the age of the CEO to various corporate policies and firm risks. Recently Serfling (2014) found that older CEOs are associated both with lower firm equity risk and with lower R&D expenditures (which are generally riskier than other firm investments), and presents evidence that CEO age is even causal in this reduction in risk. Further, several other articles have found that older CEOs are associated with lower R&D expenditures.¹ Establishing that older CEOs possess these biases is valuable, but what is not established is whether these biases tend to be beneficial or detrimental to the firm. It may be that a reduction in firm investment policy risk is precisely the reason older CEOs tend to be brought in to firms. Alternatively, it may be that this reduction in risk is deleterious to the profit potential of the firm and that the lower risk of older CEOs is the price to

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¹See, for example, Lundstrom (2002) and Levi *et al.* (2010).

be paid for their experience and ability to generate profit for the firm in other ways. This would imply that efforts to mitigate that price should be actively undertaken.

We investigate the nature of these findings, and discover several important associations that suggest that the lower risk associated with older CEOs is indeed detrimental to the firm. Specifically, we find that not only are older CEOs associated with lower R&D expenditures, but also that they distort the relationship between growth opportunities and R&D expenditures, which implies suboptimal profitability for the firm (see Hayashi, 1982 and Hubbard, 1998 for theoretical models describing the relationship between growth options and firm investment). Specifically, for older CEOs, an increase in the firm's growth options produces less R&D investment than is the case for younger CEOs.

Extending Serfling (2014), we also find that R&D expenditures appear to be the channel by which the age of the CEO is associated with firm equity risk. When a variable interacting CEO age with R&D expenditures is included in regressions explaining firm equity risk, the interaction variable is statistically and economically significant. This effect is important. For example, we will show that for a 40-year-old CEO, a one standard deviation increase in R&D equates to a 15.70% increase in systematic risk for the firm. However, for a 65-year-old CEO, the same R&D increase is associated with a 4% increase in the systematic risk of the firm. However, when we control for this interaction, direct associations between CEO age and equity risk are no longer significant. This is in contrast to Serfling (2014) who report a direct negative relationship between CEO age and equity risk of the firms.

Our findings are important to the understanding of firm management, investment, and risk, but perhaps most importantly as a corporate governance issue. Chauvin and Hirschey (1993) found that R&D expenditures have a large and positive effect on the value of the firm. Further, Eberhart *et al.* (2004) find that unexpected R&D increases result in positive abnormal returns. Suboptimal R&D expenditures are likely to hurt firm value. Consistent with this, we find that good corporate governance eliminates the distortion that age creates in the sensitivity of R&D to growth options. We measure "good corporate governance" in a variety of ways—through the use of an earnings management measure, through the use of a free cash flow measure, and by using the Herfindahl Index. In all cases, the distortion of CEO age in the sensitivity of R&D to growth options is absent among well-governed firms. This contrasts importantly with the conclusion of Serfling (2014), who asserts that firms may want older CEOs to take fewer risks.

Further, we see evidence of an indirect relationship between the age of the CEO and firm risk that relates to R&D expenditures. While Serfling (2014) found that older CEOs reduce total and idiosyncratic firm risk, we find that for well-governed firms, CEO age does not affect the systematic or idiosyncratic risk of the firm,

either directly or indirectly. In contrast, for poorly governed firms, CEO age affects the systematic and idiosyncratic risk through the channel of R&D.²

Numerous studies tie CEO characteristics and compensation to the behavior of the firm. For example, using a sample of US firms between 2005 and 2008, Huang *et al.* (2012) found a positive association between CEO age and reporting quality, and a negative association with CEO age and firms' ability to meet or beat analyst earnings expectations. Hirshleifer *et al.* (2012) found that overconfidence in managers is tied to innovation for a sample of US firms. Using a sample of S&P 500 firms, Yim (2013) found that firms governed by young CEOs are more inclined to undertake corporate acquisitions.

Empirical studies related to the investment choices of older and younger CEOs are widespread. In an influential study, Dechow and Sloan (1991) found that R&D expenditures are significantly lower in a CEO's final years of employment. Consistent with the managerial myopia conclusions of Barker and Mueller (2002), Lundstrom (2002) found that older CEOs invest less in R&D. Naveen (2006) found that R&D expenditures are negatively associated with CEO tenure. Cheng (2004), however, found that boards are effective in designing compensation schemes for CEOs that reduce the incentives to suboptimally allocate R&D expenditures. This important finding provides an indication of the endogenous relationship between CEO pay and investment levels. Hirshleifer and Thakor (1992) found that managers at the end of their career are more conservative, and that this conservatism may be good for the firm. Kini and Williams (2012) found that tournament incentives drive executives to take on greater firm risk.

That the characteristic of CEO age influences R&D expenditures in particular, and firm risk in general, is well documented. Although the empirics put forth in recent articles suggest that older CEOs tend to pursue less risky policies, there are several competing theories. As is well described in Serfling (2014), one such set of theories is related to managerial career concerns. For example, Scharfstein and Stein (1990) and Holmstrom (1999) argue that younger managers are concerned they may face greater labor market challenges if they make a bad decision early in their careers, and thus are induced to be more risk-averse. However, several other theories have been put forth as to why older CEOs may be more risk-averse. These explanations include the myopia of older CEOs (Barker and Mueller, 2002), and a desire to limit the risk of retirement benefits (Kalyta, 2009). Prendergast and Stole (1996) suggest that young executives try to signal their quality to the market with risky policies.

While there is a host of literature that ties the age or experience of the CEO of a firm to R&D, there is less work that ties R&D to the risk of the firm. Nonetheless, this relationship is established. For example, Ho *et al.* (2004) find that R&D is

²Unreported, we also find that our results are robust to recent findings that overconfidence is related to innovation (Hirshleifer *et al.*, 2012). When we control for the overconfidence measure of Malmendier and Tate (2005), our findings continue to hold.

positively related to systematic risk. Similarly, Coles *et al.* (2006) effectively tie R&D to total firm risk. Serfling (2014) comprehensively explores the issue.

There are few studies that concern the interaction of CEO age and firm risk. Child (1974) argues that the reduced firm risk (measured by lower volatility of sales growth) of older CEOs reflects the mental and physical limitations of these CEOs. Hart and Mellors (1970) also argue that the “variance of growth of the companies with ‘young’ chairman is greater than that of the other companies.” Neither of these studies measures the risk of the firm using modern methods. Serfling (2014), however, demonstrates that firm equity risk is lower for older CEOs using modern methods.

We aim to contribute to the literature in two ways. First, by demonstrating the manner in which CEO age is associated with the riskiness of the firm: both through R&D investment policy and as seen in the firm’s equity risk. In the former case, we demonstrate that CEO age distorts the relationship between growth options and R&D expenditures. In the latter case, we see that CEO age interacts with R&D spending to affect firm equity risk. We further find that when we control for this interaction, the direct effect seen so prominently in Serfling (2014) is no longer significant. Second, by demonstrating that these effects are absent when firms are well governed. Firms with effective governance do not exhibit these relationships with CEO age, while those with lesser corporate governance do. Neither the direct effect of CEO age seen in Serfling (2014) nor the indirect effect of CEO age affecting risk through the channel of R&D is found to be significant when we control for firm transparency. This appears to indicate that the effects of CEO age on firm risk are suboptimal for the firm, and that good corporate governance corrects this deficiency, contrary to previous literature.

This paper proceeds as follows. In Section 2 we outline our sample and define the relevant variables in our study. In Section 3 we present our methodology and results. In Section 4 we examine the robustness of our findings, and Section 5 concludes the paper.

2. Data

The time period of our analysis throughout the paper is 1993–2013. We collected CEO age from COMPUSTAT’s ExecuComp database. We denote this variable *CEO_AGE*. Many of the characteristics unrelated directly to age for which *CEO_AGE* may proxy are also effectively proxied by the length of a CEO’s tenure at a firm. Because of this, we control for tenure in the bulk of our analysis. We calculated the variable *TENURE* after we collected the year the CEO assumed office at the firm, also from Compustat’s ExecuComp. *TENURE* is the number of years the CEO has spent in office.

The firms in our sample consist of all NYSE, AMEX, and Nasdaq firms present in the Execucomp database. We exclude financial services firms and utility firms (SIC codes 6000–6999 and 4900–4999, respectively), as well as firms with assets of

<\$10 million and sales of <\$5 million. For all our accounting variables, including capital expenditures, R&D, and the book value of assets, we rely on COMPUSTAT through WRDS. We exclude firms with incomplete COMPUSTAT asset or sales data.

Firm size, *SIZE*, is the total assets of the firm, in millions of dollars. Edmans (2011) found that capital structure is related to the managerial decision horizon, and this may affect our results. So, we include several variables to control for capital constraints. We define financial slack (*SLACK*) as in Cleary (1999).³ *LEVERAGE* is defined as the sum of long-term debt and short-term debt divided by total assets. *LOG_SALES* is the natural log of the annual sales of the firm. Asset tangibility (*TANGIBILITY*) is defined as the book value of a firm's net fixed capital divided by the total assets. Dividend payout ratio (*DIV_RATIO*) is defined as the cash dividend divided by net income. *SALES_GROWTH* is the year-over-year percentage growth in sales of the firm. *CASH_FLOW* is the sum of earnings before extraordinary items plus depreciation, divided by property, plant, and equipment (*PP&E*). *CASH_CAPITAL* is the sum of cash and short-term investments scaled by total assets.

Our investment variables are measures of capital expenditures and research and development. *CAPEX* is capital expenditures divided by *PP&E*. Our primary measure for research and development undertaken by a firm, *R&D*, is research and development expenditures divided by the total assets of the firm.⁴

The relationship between managerial compensation and risk taking is well known.⁵ We therefore use two CEO compensation variables as controls in our regressions. *BONUS* is the CEO bonus for the year, divided by total current compensation. *CEO_EQUITY* is the percentage of the company owned by the CEO. *CASH_COMPENSATION* is the sum of the CEO's salary and bonus. We define the variables *DELTA* and *VEGA*, which are the sensitivity of CEO pay to equity price and volatility changes, respectively, as in Guay (1999) and Core and Guay (1999).

Fink *et al.* (2010) found firm maturity to be appropriately measured as the time in years since the founding of the firm. We define *FIRM_AGE* in this way. Our firm age data are the Fink *et al.* sample, which includes data through 2006, augmented by data from Jay Ritter's publicly available IPO data website.⁶

³Specifically, slack is defined as $(\text{Cash and Short-term Investments} + 0.7(\text{Total Receivables}) + 0.5(\text{Total Inventories}) - \text{Long-term Debt Due in One Year}) / (\text{Property, Plant, and Equipment})$.

⁴We use two other measures of risky investments, diversifying acquisitions and operating leverage (see Serfling, 2014 for details of these two measurements). All our results are robust with these two different measures of risky investments. These results are not reported for the sake of brevity but are available upon request. We thank an anonymous referee for suggesting these alternative measures.

⁵See, for example, DeFusco *et al.* (1990), Guay (1999), Coles *et al.* (2006), DeYoung *et al.* (2010), and Gormley and Matsa (2011).

⁶The Ritter data may be found at <http://bear.warrington.ufl.edu/ritter/FoundingDates.htm>.

For our risk variables, we use estimates of firm-level idiosyncratic and systematic volatility. We collect daily firm data from the Center for Research in Securities Prices (CRSP) via WRDS. For each firm i , in each year τ , we estimate the following five-factor model:

$$r_{it} - r_{ft} = \alpha + \beta_{1\tau i}(r_{mt} - r_{ft}) + \beta_{2\tau i}SMB_t + \beta_{3\tau i}HML + \beta_{4\tau i}UMD_t + \beta_{5\tau i}LIQ_t + \varepsilon_{it} \quad (1)$$

where r_{it} is the return to firm i at trading day t ; r_{ft} is the risk-free rate at time t , taken from 1-month expiring Treasury bills; r_{mt} is the market return, measured by the value-weighted CRSP return; SMB and HML are the small-minus-big and high-minus-low factors of Fama and French (1992), Fama and French (1993); UMD is a momentum factor, in the spirit of Carhart (1997); and LIQ is the liquidity factor of Pastor and Stambaugh (2003).⁷ We exclude firms that have fewer than 100 trading days in a given year.

For systematic volatility, our estimate for firm i in year τ is

$$SYSVOL_{it} = \ln(\sqrt{N_{it}}\hat{\sigma}_{it}^S), \quad (2)$$

where $\hat{\sigma}_{it}^S$ is the standard deviation of the predicted return to firm i in year τ , given the coefficients estimated in equation (1).

Our estimate for the idiosyncratic volatility of firm i for year τ is

$$IVOL_{it} = \ln(\sqrt{N_{it}}\hat{\sigma}_{it}^\varepsilon), \quad (3)$$

where N_{it} is the number of daily returns of firm i in year τ , and $\hat{\sigma}_{it}^\varepsilon$ is the standard deviation of ε for firm i in year τ estimated from equation (1). Finally, we define

$$TOT_VOL_{it} = \ln(\sqrt{N_{it}}\hat{\sigma}_{it}^R) \quad (4)$$

$\hat{\sigma}_{it}^R$ is the standard deviation of the excess return to the stock, $r_{it} - r_{ft}$, for firm i in year τ .

3. Methodology and Results

Descriptive statistics for our variables are provided in Table 1.

3.1. CEO Age and Investment Expenditures

We test the relationship between firm investment and *CEO_AGE* while controlling for a range of firm characteristics, and, crucially, the growth opportunities available to the firm as measured by Tobin's q . In this sense, we follow Edmans (2009) who

⁷The Pastor-Stambaugh liquidity factor is only available monthly. We therefore include the factor, but it is held constant for a given month. The other factors are available daily.

Table 1 Descriptive statistics

The sample includes firms for which data are available in ExecuComp and Compustat, excluding financials, utilities, and firms with total assets <\$10 million and sales <\$5 million. The sample firms are from years 1993 to 2013. *CEO_AGE* is the age of the CEO, as reported by ExecuComp. All accounting variables are from Compustat. *SIZE* is total assets as reported in Compustat. *SLACK* is cash and short-term CAPEX +.7(total receivables)+.5(total inventories)–long-term debt due in one year)/(property, plant, and equipment). *PP&E* is as reported in Compustat. *BOOK LEVERAGE* is defined as the sum of long-term debt and short-term debt divided by total assets. *TANGIBILITY* is defined as the book value of a firm's net fixed capital divided by the total assets. *DIV_RATIO* is defined as the cash dividend paid divided by net income. *SALES_GROWTH* is the year-over-year percentage growth in sales of the firm. *FIRM_AGE* is defined as the number of years since the firm's founding. *CAPEX* is capital expenditure normalized by *PP&E* at the beginning of the year. *R&D* is normalized by lag of assets. *CASH_FLOW* is the sum of earnings before extraordinary items and depreciation, divided by *PP&E*. *DELTA* is defined as the sensitivity of the option value with respect to a 1% change in stock price. *VEGA* is the sensitivity of the option value with respect to a 1% change in stock volatility. We use the methodology advocated by Core and Guay (1999) to calculate *DELTA* and *VEGA*. *TENURE* is the number of years the CEO is in office. *CASH CAPITAL* is calculated as the sum of cash and short-term investments scaled by total assets. *SYSTEMATIC VOLATILITY* is the standard deviation of the predicted return of a firm, using a five-factor model and daily returns. *IDIOSYNCRATIC VOLATILITY* is the standard deviation of a five-factor model using daily returns. *TOTAL_VOLATILITY* is the log of volatility using daily returns. *ROA* is defined as operating income before depreciation scaled by total assets. *CEO CASH COMPENSATION* is the sum of the bonus and salary of the CEO. *q* is the ratio of the market value of assets to the book value of assets, where the market value of total assets is the sum of the book value of total assets and the market value of equity minus the sum of the book value of common equity and the balance sheet value of deferred taxes. Market value of equity is the product of number of shares outstanding and price per share at the fiscal year end. *BONUS* is the CEO bonus for the year, divided by total current compensation. Total current compensation is the variable *tdc1* in ExecuComp. *CEO_EQUITY* is the percentage of the company owned by the CEO. *SALE* is the annual sale as reported by Compustat.

Variable	Mean	Median	SD	Lower quartile	Upper quartile	Skewness
Panel A: CEO-level variables						
<i>CEO_AGE</i>	55.603	56.000	7.512	51.000	60.000	0.256
<i>BONUS</i>	25.380	19.731	26.812	0.000	49.047	0.524
<i>TENURE</i>	6.906	5.000	7.635	2.000	10.000	1.684
<i>CEO_EQUITY</i>	2.542	0.358	6.533	0.103	1.440	5.826
<i>DELTA</i> (in 000s)	393.874	75.972	1232.000	0.000	454.623	26.390
<i>VEGA</i> (in 000s)	49.583	0.000	116.731	0.000	45.973	8.344
<i>CEO CASH COMPENSATION</i> (in 000s)	1193.284	855.000	1601.000	565.700	1287.000	14.593
Panel B: Firm-level variables						
<i>SIZE</i>	5500.606	1176.500	18 300.000	449.590	3645.500	10.936
<i>SLACK</i>	7.219	1.355	70.493	0.440	3.953	54.891
<i>BOOK LEVERAGE</i>	0.227	0.205	0.245	0.057	0.340	19.722
<i>TANGIBILITY</i>	0.279	0.210	0.234	0.090	0.417	0.905
<i>DIV_RATIO</i>	0.087	0.032	3.887	0.000	0.184	–36.543
<i>SALES_GROWTH</i>	0.144	0.080	0.942	–0.005	0.192	58.599
<i>CAPEX</i>	0.287	0.205	0.270	0.124	0.348	2.599

Table 1 (Continued)

Variable	Mean	Median	SD	Lower quartile	Upper quartile	Skewness
<i>R&D</i>	0.033	0.004	0.065	0.000	0.036	2.915
<i>CASH_FLOW</i>	0.675	0.411	1.955	0.154	0.958	1.047
<i>CASH CAPITAL</i>	383.131	41.973	5849.000	7.758	198.712	117.052
<i>PP&E</i>	1951.665	260.532	6987.000	69.103	1047.800	12.084
<i>FIRM_AGE</i>	38.799	27.000	33.857	15.000	51.000	1.673
<i>LOG_SALES</i>	7.168	7.085	1.625	6.110	8.200	−0.002
<i>SYSTEMATIC VOLATILITY</i>	0.066	0.032	0.102	0.014	0.071	3.909
<i>IDIOSYNCRATIC VOLATILITY</i>	0.039	0.034	0.021	0.025	0.048	2.350
<i>TOTAL_ VOLATILITY</i>	0.105	0.063	0.138	0.034	0.123	5.485
<i>q</i>	1.510	1.104	1.377	0.701	1.830	2.379
<i>ROA</i>	3.265	4.607	20.235	1.368	8.450	−13.357

demonstrates that investment is increasing in productivity. If our hypothesis is correct and CEO age is related to the manner in which CEOs take risks, the association between investment and q will be distorted. To model this possible distortion, we include an interaction variable of q with *CEO_AGE*. If older CEOs invest less in an absolute sense, that will be manifested in a negative coefficient on CEO age. This is the finding of Lundstrom (2002). If the age of the CEO actually distorts the relationship between investment and growth opportunities, then we will see a nonzero coefficient on the interaction between *CEO_AGE* and q .

Our specific panel data regression is

$$R\&D_{it} = \alpha_i + \kappa_i + \mu_t + \beta_1 q_{it} + \beta_2 (CEO_AGE_{it}) + \beta_3 (CEO_AGE_{it} * q_{it}) + B(\text{controls}) + \varepsilon_{it} \quad (5)$$

where q is a proxy for Tobin's q , and we follow Kaplan and Zingales (1997) in using the market-to-book value of assets for this proxy. While this measure is just a proxy for Tobin's q , Chung and Pruitt (1994) demonstrate that market-to-book-based measures for Tobin's q perform similarly to other metrics that may be used, and comprise the vast majority of the variation in more accurate measures. Since noise in this measure biases our results against statistical significance, our findings may be expected to be stronger if a more effective estimate of Tobin's q were available.

CEO_AGE is the age of the CEO of firm i at time t , B is a vector of coefficients, and controls represents our control variables. α , κ and μ are firm, industry, and year fixed effects, respectively. Even though we include a host of control variables, it may be possible that our results could be affected by unobserved, omitted, firm-

specific factors which may be correlated with *CEO_AGE*. We therefore include firm fixed effects to ensure the results are not affected by omitted time invariant factors correlated with CEO age.

Further, there may be unobserved omitted industry specific factors which may be correlated with CEO age affecting our results. In order to address this issue, we control for industry fixed effects using the two-digit SIC code.⁸

Finally, the sensitivity of investment expenditures to q may be affected by macro-level growth opportunities that are different in different years. We include year fixed effects in the regression control for such time-sensitive effects.

We include a comprehensive set of control variables. We control for firm demographic variables including firm size and age. *TENURE* proxies for a range of variables such as CEO experience, the CEO's network, and so forth. Inclusion of this variable helps strengthen *CEO_AGE* as a true instrument for the calendar age of the CEO, as it controls for many of the characteristics for which *CEO_AGE* may otherwise proxy. Edmans (2011) finds that capital constraints may induce external monitoring, and so we control for the financial condition of the firm by controlling for financial slack, firm leverage, and dividend payout ratio. Further, we control for a firm's asset tangibility and sales growth. Given the potential relationship of CEO bonuses and stock ownership with investment behavior, we also include these variables in the regression. Tobin's q , *CEO_AGE*, and all other variables in the panel regressions are standardized, to facilitate the ease of interpretation.

We use White (1980) standard errors in our regressions to account for heteroskedasticity. Further, it may be possible that observations of a firm across time are correlated with each other. The violation of the independence assumption may lead to biased standard errors. We therefore cluster the standard errors by firm to control for the possibility of the residuals being correlated across time.

We see in Table 2, Model 1 that *CEO_AGE* does have important associations with research and development investment. *CEO_AGE* is negatively associated with *R&D*, which is in line with Lundstrom (2002) and Serfling (2014). Importantly, however, we see in Model 2 that *CEO_AGE* inserts a distortion between *R&D* and growth opportunities, as proxied by Tobin's q . The coefficient on the interaction between *CEO_AGE* and Tobin's q is negative, and statistically significant at the 1% level. The coefficient on this interaction is -0.001 , while the coefficient on Tobin's q is 0.010 . A one standard deviation increase in the age of the CEO (the variable is standardized, but for reference, a one standard deviation increase in *CEO_AGE* is an increase of about seven and a half years) reduces the sensitivity of capital expenditures to Tobin's q by 10.00% ($-0.001/0.010$). In summary, we find that a unit increase in growth options for the firm results in less R&D investment for older managers than younger ones. Further, this is an economically meaningful reduction.

⁸Capital expenditure and R&D spending patterns certainly vary by industry (see, for example, Ganuza *et al.* (2009) for a discussion of R&D spending in the pharmaceutical industry).

Table 2 Sensitivity of R&D expenditures to q

Sample firms are from years 1993 to 2013. *CEO_AGE* is the age of the CEO. *CAPEX* is capital expenditures deflated by PP&E at the beginning of the fiscal year. *R&D* is deflated by lag assets. q is the market value of assets divided by the book value of assets. *SIZE* is total assets. *SLACK* is cash and short-term investments $+7(\text{Total Receivables})+5(\text{Total Inventories})-\text{long-term debt due in one year}/(\text{property, plant, and equipment})$. *BOOK LEVERAGE* is the sum of long-term debt and short-term debt divided by total assets. *TANGIBILITY* is the book value of a firm's net fixed capital divided by the total assets. *DIV_RATIO* is cash dividend paid divided by net income. *SALES_GROWTH* is the year-over-year percentage growth in sales for the firm. *FIRM_AGE* is defined as the number of years since firm founding. *BONUS* is the CEO bonus for the year, divided by total compensation. *CEO_EQUITY* is the percentage of the company owned by the CEO. *CASH_FLOW* is earnings before extraordinary items and depreciation, divided by *PP&E*. All variables are standardized. Regressions are estimated with firm fixed effect, year fixed effect, industry fixed effect (two-digit SIC code). The standard errors are robust to heteroskedasticity and clustered across firms. *t*-statistics are provided in parentheses. Asterisks indicate the level of significance: *significant at 10%, **significant at 5%, ***significant at 1%.

Dependent variable	R&D	
	[1]	[2]
Intercept	0.013 (1.85)*	0.013 (1.81)*
q	0.010 (10.23)***	0.010 (9.75)***
<i>CASH_FLOW</i>	-0.006 (-5.15)***	-0.006 (-5.13)***
<i>SIZE</i>	-0.010 (-6.49)***	-0.010 (-6.48)***
<i>TANGIBILITY</i>	-0.004 (-3.40)***	-0.004 (-3.42)***
<i>BOOK LEVERAGE</i>	0.000 (-0.28)	0.000 (-0.25)
<i>DIV_RATIO</i>	0.000 (-1.41)	0.000 (-1.41)
<i>FIRM_AGE</i>	-0.001 (-2.70)***	-0.001 (-2.59)***
<i>SLACK</i>	0.000 (4.19)***	0.000 (4.00)***
<i>SALES_GROWTH</i>	0.002 (2.34)**	0.002 (2.34)**
<i>BONUS</i>	0.002 (3.90)***	0.002 (3.99)***
<i>CEO_EQUITY</i>	0.000 (-1.37)	-0.001 (-1.47)
<i>TENURE</i>	-0.001 (-0.98)	0.000 (-0.93)
<i>CEO_AGE</i>	-0.001 (-1.88)*	-0.001 (-2.04)**
<i>CEO_AGE</i> * q		-0.001 (-2.27)**
<i>N</i>	21 293	21 293
<i>R</i> ²	0.434	0.434

3.2. Firm Risk, R&D Expenditures, and CEO Age

We estimate the relationship between systematic, idiosyncratic, and total risk (defined in equations 2–4, respectively) and CEO age with a panel regression. Our approach here follows Coles *et al.* (2006), though we include a slightly more comprehensive set of control variables, as well as fixed effects for the firm, industry, and year for each observation.

One set of variables that provides crucial controls in our regression is related to the compensation scheme of CEOs. Guay (1999) found that firms use compensation to encourage appropriate CEO risk taking. We therefore include a measure of the delta and vega of the CEO compensation as control variables. In Section 2, we

define the control variables *DELTA* and *VEGA*. However, direct inclusion of *DELTA* and *VEGA* as independent variables may give rise to an endogeneity problem in our panel regressions. For example, Cheng (2004) found that corporate boards are effective in constructing pay packages for CEOs to mitigate incentives for inappropriate risk-taking behavior. To avoid this endogeneity issue, we estimate the following regressions:

$$\begin{aligned} DELTA_{it} = & \eta_0 + \eta_1(TOT_VOL_{it}) + \eta_2(VEGA_{it}) + \eta_3(TENURE_{it}) + \eta_4(q_{it}) \\ & + \eta_5(CASH_{it}) + \eta_6(R\&D_{it}) + \eta_7(CAPEX_{it}) + \eta_8(LOG_SALE_{it}) \\ & + \eta_9(LEVERAGE_{it}) + \varepsilon_D \end{aligned} \quad (6)$$

$$\begin{aligned} VEGA_{it} = & \eta_0 + \eta_1(TOT_RISK_{it}) + \eta_2(DELTA_{it}) + \eta_3(CASH_COMPENSATION_{it}) \\ & + \eta_4(q_{it}) + \eta_5(R\&D_{it}) + \eta_6(CAPEX_{it}) + \eta_7(LOG_SALE_{it}) \\ & + \eta_8(LEVERAGE_{it}) + \varepsilon_V \end{aligned} \quad (7)$$

where i indexes firms and t indexes years. The specification of these regressions is the same as that of Coles *et al.* (2006), and we further follow their methodology by using the residuals ε_D and ε_V with a lag in our panel regressions. By using these lagged residuals, we remove the effect of firm risk on *DELTA* and *VEGA*, and hence control for endogeneity.

Columns 1, 3, and 5 of Table 3 provide the parameter estimates for our initial set of panel regressions, which exclude the interaction of *CEO_AGE* with R&D expenditures, in line with previous research. As can be seen in Table 3, *CEO_AGE* has a negative relationship with the three different measures of risk. This result is in line with Serfling (2014). The coefficients on the control variables largely align with intuition. In line with Cao *et al.* (2008), Tobin's q is positively associated with idiosyncratic risk, as well as systematic risk. *R&D* is positive and significantly associated with all three measures of risk.

Columns 2, 4, and 6 of Table 3 provide the same regression, but include the interaction between *CEO_AGE* and *R&D* as an independent variable (we also include interaction between *CAPEX* and *CEO_AGE* for completeness). This interaction is negative and significant at the 1% level for all three risk measures. This negative value for the interaction term implies that the increase in risk due to R&D spending is less, the older the CEO. CEO age is clearly associated with risk through the channel of R&D spending.

Our findings in Table 3 indicate that *CEO_AGE* is a determinant of the relationship between *R&D* and firm risk. This is an economically important effect. Consider the percentage change in total risk, given the coefficients in Model 2, of a one standard deviation increase in *R&D* (0.065). This percentage change is: $3.18(0.065) - 0.046(CEO_AGE)(0.065)$. For a 40-year-old CEO, the one standard deviation increase in *R&D* equates to a 8.71% increase in total risk for the firm.

Table 3 Sensitivity of firm risk on CEO age and $R\&D$

The sample includes firms for which data are available in ExecuComp and Compustat, excluding financials, utilities, and firms with total assets <\$10 million and sales <\$5 million. The sample firms are from years 1993 to 2013. CEO_AGE is from ExecuComp. All accounting variables are from Compustat. CEO_AGE is the age of the CEO, as reported by ExecuComp. $SIZE$ is the total assets as reported in Compustat. $BOOK_LEVERAGE$ is defined as the sum of long-term debt and short-term debt divided by total assets. $CAPEX$ is capital expenditure normalized by $PP\&E$ at the beginning of the year. $R\&D$ is normalized by lag of assets. $SYSVOL$ is the standard deviation of the predicted return of a firm, using a five-factor model and daily returns. $IVOL$ is the standard deviation of the residual from a five-factor model using daily returns. TOT_VOL is defined as in equation (4). $DELTA$ is defined as the sensitivity of the option value with respect to a 1% change in stock price. $VEGA$ is the sensitivity of the option value with respect to a 1% change in stock volatility. We use the methodology advocated by Core and Guay (1999) to calculate $DELTA$ and $VEGA$. Residual $DELTA$ is the residual of the regression estimate of $DELTA$ on $SYSTEMATIC_VOLATILITY$ and $IDIOSYNCRATIC_VOLATILITY$ respectively and other control variables as defined by equation (4). Residual $VEGA$ is the residual of the regression estimate of $VEGA$ on $SYSTEMATIC_VOLATILITY$ and $IDIOSYNCRATIC_VOLATILITY$ respectively and other control variables as defined by equation (5). $TENURE$ is the number of years the CEO is in office. $CASH_CAPITAL$ is calculated as the sum of cash and short-term investments scaled by total assets. $CEO_CASH_COMPENSATION$ is the sum of the bonus and salary of the CEO. ROA is defined as operating income before depreciation scaled by total assets. q is the ratio of the market value of assets to the book value of assets, where the market value of total assets is the sum of the book value of total assets and the market value of equity minus the sum of the book value of common equity and the balance sheet value of deferred taxes. The market value of equity is the product of number of shares outstanding and price per share at the fiscal year end. LOG_SALE is the logarithm of $SALE$, where $SALE$ is the annual sale as reported by Compustat. Regressions are estimated with firm fixed effect, year fixed effect, industry fixed effect (two-digit SIC code). The standard errors are robust to heteroskedasticity and clustered across firms. t -statistics are provided in parentheses. Asterisks indicate the level of significance: *significant at 10%, **significant at 5%, ***significant at 1%.

Dependent variable	Log(TOT_VOL)			Log($SYSVOL$)			Log($IVOL$)		
	[1]	[2]	[3]	[4]	[5]	[6]	[7]	[8]	[9]
Constant	-1.808 (-20.66)***	-1.977 (-19.54)***	-3.265 (-28.81)***	-3.507 (-28.31)***	6.657 (149.48)***	6.587 (125.47)***			
Lag residual $VEGA$	0.060 (1.15)	0.065 (1.26)	0.109 (1.19)	0.117 (1.29)	0.017 (0.74)	0.019 (0.83)			
Lag residual $DELTA$	-0.007 (-1.46)	-0.007 (-1.44)	-0.010 (-1.40)	-0.010 (-1.35)	-0.002 (-0.89)	-0.002 (-0.87)			
$TENURE$	0.003 (2.19)**	0.002 (1.93)*	0.004 (2.70)***	0.004 (2.42)***	0.001 (1.60)	0.001 (1.39)			
CEO_CASH	0.000 (-1.04)	0.000 (-0.99)	0.000 (0.85)	0.000 (0.97)	0.000 (-1.19)	0.000 (-1.16)			
$COMPENSATION$									
$SIZE$	-0.166 (-21.13)***	-0.166 (-21.27)***	0.000 (0.02)	-0.001 (-0.07)	-0.105 (-26.95)***	-0.105 (-27.05)***			
$SALES_GROWTH$	0.000 (2.46)**	0.000 (2.47)**	0.000 (2.40)**	0.000 (2.37)**	0.000 (3.09)***	0.000 (3.09)***			
$R\&D$	0.805 (5.32)***	3.180 (4.03)***	1.526 (7.02)***	5.295 (5.18)***	0.391 (5.47)***	1.329 (3.53)***			

Table 3 (Continued)

Dependent variable	Log(TOT_VOL)			Log(SYSVOL)			Log(IVOL)		
	[1]	[2]	[3]	[4]	[5]	[6]	[5]	[6]	[6]
CAPEX	0.128 (5.11)***	0.392 (2.81)***	0.296 (8.74)***	0.624 (3.06)***	0.048 (3.88)***	0.163 (2.34)**			
<i>q</i>	0.033 (5.85)***	0.033 (5.92)***	0.066 (8.68)***	0.066 (8.81)***	0.009 (3.24)***	0.009 (3.28)***			
BOOK LEVERAGE	0.304 (4.73)***	0.305 (4.84)***	-0.053 (-0.71)	-0.050 (-0.69)	0.192 (5.82)***	0.193 (5.92)***			
ROA	-0.008 (-8.42)***	-0.008 (-8.49)***	-0.005 (-8.00)***	-0.005 (-8.23)***	-0.004 (-7.94)***	-0.004 (-7.99)***			
FIRM_AGE	-0.003 (-9.30)***	-0.003 (9.32)***	-0.001 (-4.47)**	-0.001 (-4.50)**	-0.002 (-10.40)***	-0.002 (-10.41)***			
CASH CAPITAL	0.000 (-5.84)***	0.000 (-6.23)***	0.000 (-2.45)**	0.000 (-2.58)**	0.000 (-6.30)***	0.000 (-6.59)***			
STOCK RETURNS	0.001 (9.57)***	0.001 (9.46)***	0.001 (6.97)***	0.001 (6.94)***	0.000 (7.35)***	0.000 (7.27)***			
CEO_AGE	-0.004 (-3.53)***	-0.001 (-0.49)	-0.004 (-2.46)**	0.001 (0.47)	-0.002 (-3.23)***	0.000 (-0.63)			
CEO_AGE*CAPEX		-0.005 (-1.98)**		-0.006 (-1.69)*		-0.002 (-1.72)			
CEO_AGE*R&D		-0.046 (-3.16)***		-0.072 (-3.88)***		-0.018 (-2.59)**			
N	19 876	19 876	19 876	19 876	19 876	19 876			
R ²	0.850	0.850	0.629	0.629	0.853	0.853			

However, for a 65-year-old CEO, the same R&D increase is associated with only a 1.23% increase in the total risk of the firm. The age of the CEO affects the total risk of the firm indirectly, through the channel of *R&D*. The interaction of *CEO_AGE* with *R&D* has a coefficient of -0.046 and represents this channel. It represents the cross-partial derivative of the log of total risk with respect to *R&D* and *CEO_AGE*. It therefore measures the effect of a unit change in *CEO_AGE* on the percentage change in total risk resulting from an increase in *R&D*. So, for example, a one standard deviation increase in *CEO_AGE* (which is 7.51 years; these variables aren't standardized because of the resulting difficulty in interpreting a dependent variable in logs) implies a 10.86% ($7.51 \times 0.046 / 3.18$) reduction in the sensitivity of total risk to *R&D*.

In Model 4, we examine the percentage change in systematic risk of a one standard deviation increase in *R&D* (0.065). This percentage change is given by $5.295(0.065) - 0.072(CEO_AGE)(0.065)$. For a 40-year-old CEO, the one standard deviation increase in *R&D* equates to a 15.70% increase in systematic risk for the firm. For a 65-year-old CEO, the same R&D increase is associated with only a 4% increase in the systematic risk of the firm. A one standard deviation increase in *CEO_AGE* implies a 10.22% ($7.51 \times 0.072 / 5.29$) reduction in the sensitivity of systematic risk to *R&D*.

Looking at Model 6, we see that while statistically significant, the -0.018 coefficient on *CEO_AGE***R&D* indicates that this interaction is considerably less economically important in describing idiosyncratic risk than systematic risk. Based on our parameter estimates in Model 6, a one standard deviation increase in R&D results in a percentage change of $1.329(0.065) - 0.018(CEO_AGE)(0.065)$. This implies that a one standard deviation increase in R&D results in a 3.95% increase in idiosyncratic risk if the CEO is 40 years old, and a 1.03% increase in idiosyncratic risk if the CEO is aged 65. The effect is still present, and even statistically significant at the 1% level, but it is considerably weaker than in describing systematic risk. We note that *CEO_AGE* becomes statistically insignificant once the interaction of *CEO_AGE* with R&D is included in the regression estimations. This result has not been previously reported in the literature and stands in sharp contrast with Serfling (2014).

The coefficient estimates in Table 3 imply that older CEOs reduce the systematic risk of the firm, but only through the channel of R&D expenditures. But why are R&D expenditures less associated with systematic risk if they are undertaken by older CEOs compared to younger ones? Is this reduction in systematic risk beneficial or harmful to the firm? It is possible that the answers to these questions lie in our analysis presented in Table 2. The partial correlation between R&D and growth opportunities is less for firms governed by older CEOs. These growth opportunities are likely (though certainly not exclusively) economic prospects of the firm that are associated with the state of the macro economy. If older CEOs choose R&D projects that are less closely affiliated with the systematic risks of the economy, we would expect to see reductions in systematic risk through the channel of R&D.

3.3. Corporate Governance, CEO Age, and Firm Risk

There exists significant heterogeneity of corporate governance mechanisms across firms. To the extent that these mechanisms are effective, we expect that they would reduce suboptimal investment policies by managers. We see in previous sections that the age of the CEO is associated with a distortion in the relationship between growth options and firm R&D investment. Previous research suggests that such a distortion is suboptimal. We therefore would expect that if corporate governance mechanisms are effective they would reduce this distortion, which according to theory would increase the present value of the firm. Similarly, if the reduction in firm risk that we have established is associated with older CEOs through the channel of R&D expenditures is harmful to the firm, we expect that firms with good corporate governance would take steps to limit this reduction.

Our goal in this section is to determine whether well-governed firms are able to mitigate the distortions in R&D expenditure policy and firm risk that appear so evidently in Tables 2 and 3.⁹ To see if this is the case, we repeat our analysis from the previous sections but divide our firms into two groups—well-governed and less well-governed firms. The quality of corporate governance for a particular firm is a rather difficult dimension to observe, and consequently there are several metrics used in the literature. For robustness, we employ here three different such metrics to divide firms into “good” and “bad” governance groups: earnings management, the coincidence of free cash flow with high and low Tobin’s q , and the Herfindahl Index.

In order to calculate the measure of earnings management, we follow the modified Jones (1991) model, which is perhaps the best available measure of earnings management (see Dechow *et al.*, 1995; Guay *et al.*, 1996; Chung *et al.*, 2002; Davidson *et al.*, 2004; Chang *et al.*, 2016). Total accruals (TAC) are divided into non-discretionary accruals ($NDAC$) and discretionary accruals (DAC).

$$\frac{TAC_{i,t}}{TA_{i,t-1}} = NDAC_{i,t} + DAC_{i,t} \quad (8)$$

Following Chung *et al.* (2002) and Dechow *et al.* (1995), the following regression is estimated for TAC , for each industry, for every year, with the industry being defined by the two digit SIC code:

$$\frac{TAC_{i,t}}{TA_{i,t-1}} = \beta_0 + \beta_1 \frac{1}{TA_{i,t-1}} + \beta_2 \left[\frac{(\Delta REV_{i,t} - \Delta AR_{i,t})}{TA_{i,t-1}} \right] + \beta_3 \frac{PPE_{i,t}}{TA_{i,t-1}} + \varepsilon_{i,t} \quad (9)$$

where i denotes the firm and t denotes the year. $\Delta REV_{i,t}$ is the change in sales revenue and $\Delta AR_{i,t}$ is the change in accounts receivable. In both cases the Δ indicates

⁹In this section, we use “well governed” and “closely monitored” as close synonyms. This is by design. In line with the literature, our implicit assumption here is that managers in firms that are more closely monitored have less opportunity to behave in ways that reduce the value of the firm but increase the well-being of the managers.

the change from time $t-1$ to t . $PPE_{i,t}$ is net property, plant, and equipment, $TA_{i,t-1}$ is total assets, and $\varepsilon_{i,t}$ is the error term.

TAC is measured by:

$$TAC_{i,t} = (\Delta CA_{i,t} - \Delta CL_{i,t} - \Delta Cash_{i,t} + \Delta STD_{i,t} - Dep_{i,t}) \quad (10)$$

where $\Delta CA_{i,t}$ is the change in current assets, $\Delta CL_{i,t}$ is the change in current liabilities, $\Delta Cash_{i,t}$ is the change in cash, $\Delta STD_{i,t}$ is the change in debt included in current liabilities, and $Dep_{i,t}$ is depreciation and amortization expense.

Equation (9) is estimated cross-sectionally for every year and every industry (two-digit SIC code). Non-discretionary accruals (NDAC) is the fitted values from the regression estimation of equation (9). The discretionary accruals (DAC) are defined as the residuals ($\varepsilon_{i,t}$) from the regression equation (9).

Discretionary accruals provide a proxy for earnings management, and as such firms with discretionary accruals more than the industry median are classified as “high earnings management” firms. These firms with high discretionary accruals are generally regarded as those with greater agency problems. Similarly, we classify firms with discretionary accruals below the industry median as the “low earnings management” firms. These firms are believed to exhibit fewer agency problems (for an excellent discussion this, see Chang *et al.*, 2016).

The coincidence of free cash flow with high and low Tobin’s q is another metric of the effectiveness of a firm’s corporate governance (see Jurkus *et al.*, 2011). Free cash flow is calculated as in Lang *et al.* (1991). Free cash flow (FCF) is calculated as operating income before depreciation, minus interest expense minus income taxes plus increases in the deferred tax and investment tax credit minus preferred dividends minus common dividends. Following Jurkus *et al.* (2011), for this metric we consider only firms with positive free cash flow. We separate firms with positive free cash flow based on Tobin’s q . Firms with Tobin’s q less than one are identified as firms with poor investment opportunity and are thus considered likely to have a greater number of residual agency issues. By contrast, firms with positive free cash flow and Tobin’s q greater than one are firms with good investment opportunities and are likely to have fewer agency issues.

As a final proxy for effective corporate governance, we employ the Herfindahl Index based on sales (also called HI).

For year t and for industry j HI based on sales is calculated as follows:

$$HI_{j,t} = \sum_i \left(\frac{SALES_{i,j,t}}{SALES_{j,t}} \right)^2 \quad (11)$$

where $SALES_{i,j,t}$ is the revenue of firm i (in industry j) in fiscal year t and $SALES_{j,t}$ is the total revenue of all firms in industry j for fiscal year t . Following Chang *et al.* (2016), we interpret firms in more competitive industries (low HI) as likely to have fewer agency problems. Firms in less competitive industries (High HI) by contrast

are likely to have a greater number of agency issues. We calculate the median HI across all industries every year. Firms with HI above the median HI are supposed to be less competitive and hence have lesser transparency and more agency problems. Similarly, firms with HI below the median HI are believed to be more competitive, and hence have fewer agency problems.

Table 4 re-estimates the regressions of Table 2, while subdividing the data in important ways to reflect different corporate governance regimes among firms. Models 1 and 2 estimate the regression dividing firm-years into low and high earnings management firms. Chung *et al.* (2005) link firms with high earnings management to opportunistic behavior among firms, and find that external monitoring reduces its practice. In Model 1 of Table 4 we see that firms with low earnings management exhibit no distortion on the part of *CEO_AGE* in the relationship between q and R&D expenditures. The coefficient on *CEO_AGE** q is insignificant at any conventional level. By contrast, the corresponding coefficient in Model 2 is significant at the 1% level, and economically meaningful. Casually speaking, “poorly governed firms” by this metric exhibit a significant breakdown in the link between R&D expenditures and growth opportunities that economic theory suggests is optimal for profitability, and *CEO_AGE* is associated with this breakdown. Meanwhile well-governed firms exhibit no such distortion at all.

We see very similar results for our other metrics of corporate governance. In Models 3 and 5 we see the respective coefficients on *CEO_AGE** q are statistically insignificant for firms with a coincidence of positive free cash flow and high Tobin’s q , and for firms in a low HI environment—both proxies for “well governed,” transparent firms. By contrast, their counterpart coefficients in Models 4 and 6 correspond to statistically and economically significant distortions of *CEO_AGE* between growth opportunities and R&D expenditures. The differences between well-governed and poorly governed firms is striking.

In a similar spirit, Table 5 re-estimates the equations in Table 3, while partitioning the data into firms with low and high degrees of earnings management. Regardless of the risk measure, *CEO_AGE***R&D* is negative and significant (at the 1% level for total and systematic volatility, and at the 5% level for idiosyncratic volatility) for firms that exhibit a high degree of earnings management. This may be seen in Models 2, 4, and 6. Meanwhile, the corresponding coefficients are statistically insignificant for firms with low earnings management, which may be seen in Models 1, 3, and 5. It is interesting to note that in these models, low earnings management firms not only exhibit no significance on the *CEO_AGE***R&D* coefficient, but the coefficient on *CEO_AGE* is insignificant as well. In addition to eliminating the indirect channel by which *CEO_AGE* affects firm risk through R&D, these better managed firms also eliminate the direct effect of *CEO_AGE* on firm risk. This result has not been reported in the literature before and potentially affects the interpretation of findings in Serfling (2014), in that it is suggestive that firms do not desire the influence of CEO age on their risk profile. Had firms hired older CEOs with the intention that they would reduce their risk, we would expect

to see significant coefficients on *CEO_AGE* and/or *CEO_AGE***R&D* for well-governed firms.

Tables 6 and 7 report analogous results to Table 5. Whether we partition the data by the coincidence of Tobin's *q* with free cash flow, or by the Herfindahl Index of sales, the results are consistent. The coefficients on *CEO_AGE***R&D* (as well as *CEO_AGE* alone) are statistically insignificant when we confine our analysis to "well-governed" firms. In Table 6, these are firms that have positive cash flow and high *q*. In Table 7, these are firms that have a low HI (meaning they face greater competition for profits, and presumably are more transparent). And yet uniformly, in both tables, "poorly governed" firms exhibit negative and statistically significant effects of *CEO_AGE* on the relationship between *R&D* and firm risk. As we concluded upon review of Table 5, these results appear to suggest that the appointment of older CEOs is not an attempt to negatively influence either systematic or idiosyncratic firm risk. If it were, we would expect well-governed, more transparent firms to retain this association rather than eliminate it. However, as we see in Tables 5–7, only less transparent, presumably more questionably governed firms exhibit this association.

4. Robustness: Propensity Score Matching

Despite the care taken in the construction of the econometric experiment in the preceding sections, there still exists the potential risk of endogeneity in our results. For example, it may be that the characteristics of firms hiring older CEOs are systematically different from those hiring younger CEOs.¹⁰ Younger firms hire younger CEOs and older firms hire mature CEOs in order to fit with the objectives of the respective firms. To address concerns that such endogeneity bias may be driving our results, we next construct a propensity score-matching procedure.

Following propensity score-matching methodology, we divide the firms into sample and control firms. We define the sample firms to be those firms with CEO age greater than or equal to 56 (the median CEO age in our sample). The control firms are firms with CEOs aged less than 56. We next create a matched control firm for each of our sample firms. We match firms in three dimensions: *q*, *FIRM_AGE*, and *SIZE*. We form deciles of the sample firms based on the values of these three variables. The largest size and largest firm age deciles are further subdivided into two subgroups to obtain a better match (to help mitigate the extreme size and age differences of firms in the top decile). Thus we sort the total sample (all sample and control firms) into 1,210 portfolios based on *SIZE* (11 portfolios), *FIRM_AGE* (11 portfolios), and *q* (10 portfolios) every year.

For every firm in the sample, we generate a propensity score. This is estimated via a logistic regression of the firm type (1 = sample, 0 = control) on the three variables of interest, namely *SIZE*, *FIRM_AGE*, and *q*. The results of this logistic regression

¹⁰We thank an anonymous referee for insight on this point.

Table 4 Sensitivity of R&D expenditures to q , by firm transparency

The sample firms are from years 1993 to 2013. All accounting variables are from Compustat. CEO_AGE is the age of the CEO, as reported by ExecuComp. Dependent variables are defined as in Table 2. Regressions are estimated with firm fixed effect, year fixed effect, industry fixed effect (two-digit SIC code). The standard errors are robust to heteroskedasticity and clustered across firms. t -statistics are provided in parentheses. For high and low earnings management firms we follow the modified Jones (1991). Firms with discretionary accruals greater than the industry median are classified as “High earnings management” firms. Similarly, firms with discretionary accruals below the industry median are classified as the “Low earnings management” firms. Firms with positive free cash flow are divided based on Tobin’s q . Firms with positive free cash flow and Tobin’s q less than 1 are classified “Positive free cash flow and low Tobin’s q .” Similarly, firms with positive free cash flow and Tobin’s q greater than 1 are classified “Positive free cash flow and high Tobin’s q .” We calculate the median HI across all industries every year. Firms with HI above the median HI are “High HI” firms. Firms below this measure are “Low HI firms.” Asterisks indicate the level of significance: *significant at 10%, **significant at 5%, ***significant at 1%.

Dependent variable	R&D					
	Low earnings management [1]	High earnings management [2]	Positive free cash flow and high Tobin’s q [3]	Positive free cash flow and low Tobin’s q [4]	Low HI [5]	High HI [6]
Intercept	0.566 (2.420)**	0.369 (9.160)***	0.069 (1.250)	0.393 (1.930)**	0.073 (16.010)***	0.117 (4.110)***
q	0.173 (3.130)***	0.051 (1.810)*	0.077 (2.440)**	0.071 (6.620)***	0.014 (1.800)*	0.256 (3.000)***
CASH_FLOW	-0.078 (-1.890)*	0.004 (0.370)	0.000 (-0.100)	0.007 (0.630)	0.003 (0.300)	-0.065 (-2.120)**
SIZE	-0.117 (-0.890)	-0.071 (-4.580)***	0.008 (0.340)	-0.044 (-1.500)	-0.003 (-0.460)	-0.255 (-2.800)***
TANGIBILITY	-0.890 (-4.350)***	-0.114 (-6.720)***	-0.023 (-2.280)**	-0.141 (-5.140)***	-0.009 (-2.650)***	-0.326 (-4.040)***
BOOK	0.200 (1.540)	-0.009 (-1.040)	0.007 (0.850)	-0.008 (-0.620)	-0.002 (-0.580)	0.134 (3.380)***
LEVERAGE						
DIV_RATIO	0.003 (1.020)	-0.004 (-0.780)	-0.001 (-0.950)	-0.002 (-0.330)	0.000 (-0.320)	-0.002 (-0.090)
FIRM_AGE	-0.036 (-0.700)	-0.005 (-0.480)	0.008 (1.540)	-0.009 (-0.500)	0.003 (1.530)	0.011 (0.190)
SLACK	-0.009 (-0.230)	0.243 (25.340)***	0.038 (2.110)**	0.012 (1.320)	0.003 (0.520)	0.000 (0.010)
SALES_	0.168 (3.730)***	0.011 (0.610)	0.031 (1.840)*	0.003 (0.390)	0.562 (1.510)	0.317 (12.150)***
GROWTH						

Table 4 (Continued)

R&D						
Dependent variable	Low earnings management [1]	High earnings management [2]	Positive free cash flow and high Tobin's q [3]	Positive free cash flow and low Tobin's q [4]	Low HI [5]	High HI [6]
BONUS	0.027 (1.090)	-0.020 (-3.210)***	0.000 (0.090)	-0.005 (-0.440)	-0.002 (-0.990)	0.002 (0.050)
CEO_EQUITY	-0.017 (-0.017)	-0.005 (-0.510)	-0.001 (-0.310)	-0.008 (-0.630)	-0.001 (-0.520)	-0.015 (-0.350)
TENURE	-0.006 (-0.130)	-0.010 (-1.070)	-0.016 (-1.510)	-0.004 (-0.310)	-0.001 (-0.420)	-0.032 (-0.690)
CEO_AGE	-0.010 (-0.210)	0.007 (0.810)	0.010 (1.060)	-0.001 (-0.090)	-0.001 (-0.400)	-0.012 (-0.280)
CEO_AGE* q	-0.054 (-0.890)	-0.074 (-3.380)***	-0.009 (-0.340)	-0.092 (-11.700)***	-0.023 (-1.490)	-0.129 (-2.310)**
N	9507	8488	10 397	5502	8252	9733
R ²	0.066	0.300	0.138	0.080	0.082	0.117

Table 5 Sensitivity of firm risk on CEO age and R&D by earnings management

The sample firms are from years 1993 to 2013. All accounting variables are from Compustat. *CEO_AGE* is the age of the CEO, as reported by ExecuComp. Independent and dependent variables are defined as in Table 3. Regressions are estimated with firm fixed effect, year fixed effect, industry fixed effect (two-digit SIC code). The standard errors are robust to heteroskedasticity and clustered across firms. *t*-statistics are provided in parentheses. For high and low earnings management firms we follow the modified Jones (1991). Firms with discretionary accruals greater than the industry median are classified as “High EM” firms. Similarly, firms with discretionary accruals below the industry median are classified as the “Low EM” firms. Asterisks indicate the level of significance: *significant at 10%, **significant at 5%, ***significant at 1%.

Dependent variable	Log(total volatility)		Log(systematic volatility)		Log(idiosyncratic volatility)	
	Low EM	High EM	Low EM	High EM	Low EM	High EM
	[1]	[2]	[3]	[4]	[5]	[6]
Constant	-5.880 (-18.060)***	-5.703 (-29.920)***	-2.153 (-4.650)***	-3.120 (-10.860)***	4.253 (25.500)***	4.500 (46.540)***
Lag residual	-0.089 (-0.760)	0.125 (1.720)*	-0.068 (-0.440)	0.232 (2.130)**	-0.080 (-1.330)	0.012 (0.320)
VEGA						
Lag residual	0.000 (-0.020)	-0.004 (-0.510)	0.008 (0.460)	-0.012 (-1.120)	-0.002 (-0.310)	-0.001 (-0.400)
DELTA						
TENURE	0.002 (1.050)	0.002 (1.740)*	0.005 (1.480)	0.004 (1.730)*	0.000 (0.260)	0.001 (1.990)**
CEO CASH	0.000 (0.760)	0.000 (-1.890)*	0.000 (1.490)	0.000 (-1.120)	0.000 (0.760)	0.000 (-1.400)
COMPENSATION						
SIZE	-0.202 (-6.830)***	-0.250 (-17.480)***	-0.086 (-2.010)**	-0.036 (-1.660)*	-0.125 (-8.990)***	-0.161 (-22.150)***
SALES GROWTH	0.024 (0.770)	0.085 (4.680)***	0.013 (0.280)	0.091 (3.320)***	0.027 (1.660)*	0.046 (5.040)***
R&D	1.499 (0.780)	3.082 (3.910)***	2.813 (0.940)	6.354 (5.350)***	0.902 (0.980)	1.145 (2.860)***
CAPEX	0.335 (0.990)	0.081 (0.480)	0.332 (0.540)	0.409 (1.600)	0.111 (0.650)	0.013 (0.160)
<i>q</i>	-0.011 (-1.160)	0.017 (3.230)***	-0.003 (-0.220)	0.031 (3.860)***	-0.011 (-2.300)**	0.003 (0.980)
BOOK LEVERAGE	0.268 (4.710)***	0.230 (5.240)***	0.000 (-0.010)	0.134 (2.020)**	0.181 (5.950)***	0.121 (5.430)***
ROA	-0.009 (-6.560)***	-0.007 (-12.650)***	-0.005 (-3.880)***	-0.002 (-2.460)**	-0.005 (-6.780)***	-0.004 (-13.240)***
FIRM_AGE	-0.001 (-1.620)	-0.001 (-1.140)	0.001 (0.880)	0.000 (0.130)	-0.001 (-2.340)**	0.000 (-0.920)
CASH CAPITAL	0.000 (-0.100)	0.000 (-1.690)*	0.000 (-0.210)	0.000 (-1.770)	0.000 (0.170)	0.000 (-1.160)
STOCK RETURNS	0.000 (6.090)***	0.000 (0.190)	0.000 (2.910)***	0.000 (-3.670)***	0.000 (5.930)***	0.000 (-0.190)

Table 5 (Continued)

Dependent variable	Log(total volatility)		Log(systematic volatility)		Log(idiosyncratic volatility)	
	Low EM [1]	High EM [2]	Low EM [3]	High EM [4]	Low EM [5]	High EM [6]
CEO_AGE	-0.001 (-0.340)	0.001 (0.560)	-0.004 (-1.070)	-0.006 (-2.070)**	0.000 (0.220)	0.000 (0.020)
CEO_AGE*CAPEX	-0.002 (-0.400)	0.001 (0.280)	0.001 (0.130)	0.000 (-0.080)	-0.001 (-0.360)	0.000 (0.170)
CEO_AGE*R&D	-0.038 (-1.140)	-0.054 (-3.660)***	-0.055 (-1.050)	-0.103 (-4.620)***	-0.022 (-1.370)	-0.020 (-2.730)***
N	7942	7045	7942	7045	7942	7045
R ²	0.509	0.449	0.464	0.406	0.490	0.454

Table 6 Sensitivity of firm risk on CEO age and R&D by positive free cash flow and Tobin's q

The sample firms are from years 1993 to 2013. All accounting variables are from Compustat. CEO_AGE is the age of the CEO, as reported by ExecuComp. Independent and dependent variables are defined as in Table 3. Regressions are estimated with firm fixed effect, year fixed effect, industry fixed effect (two-digit SIC code). The standard errors are robust to heteroskedasticity and clustered across firms. t -statistics are provided in parentheses. Firms with positive free cash flow are divided based on Tobin's q . Firms with positive free cash flow and Tobin's q less than 1 are classified "PCF low q ." Similarly, firms with positive free cash flow and Tobin's q greater than 1 are classified "PCF high q ." Asterisks indicate the level of significance: *significant at 10%, **significant at 5%, ***significant at 1%.

Dependent variable	Log(total volatility)			Log(systematic volatility)			Log(idiosyncratic volatility)		
	PCF high q [1]	PCF low q [2]		PCF high q [3]	PCF low q [4]		PCF high q [5]	PCF low q [6]	
Constant	-5.373 (-33.480)***	-6.804 (-44.300)***		-2.774 (-13.380)***	-4.719 (-11.920)***		4.629 (55.330)***	4.063 (52.600)***	
Lag residual	-0.219 (-1.440)	0.021 (0.380)		-0.153 (-0.880)	0.131 (1.250)		-0.218 (-2.680)***	-0.036 (-1.270)	
VEGA									
Lag residual	-0.039 (-1.860)*	0.001 (0.160)		-0.044 (-1.810)*	-0.005		-0.020 (-1.860)*	0.000 (-0.090)	
DELTA									
TENURE	0.000 (0.030)	0.003 (2.650)***		0.000 (0.100)	0.006 (2.090)**		0.000 (-0.380)	0.001 (1.630)	
CEO CASH	0.000 (1.450)	0.000 (-2.160)**		0.000 (1.660)*	0.000 (-0.360)		0.000 (1.250)	0.000 (-1.830)*	
COMPENSATION									
SIZE	-0.189 (-16.030)***	-0.180 (-14.470)***		0.046 (2.790)***	-0.009 (-0.260)		-0.125 (-20.420)***	-0.115 (-18.410)***	
SALES GROWTH	0.058 (2.440)**	0.060 (3.620)***		0.003 (0.090)	0.072 (2.460)**		0.045 (3.580)***	0.030 (3.590)***	
R&D	2.477 (1.310)	1.794 (2.750)***		5.381 (1.820)*	4.311 (2.780)***		1.209 (1.370)	0.557 (1.690)*	
CAPEX	-0.093 (-0.260)	0.195 (1.350)		-0.495 (-0.810)	0.356 (1.240)		-0.019 (-0.100)	0.080 (1.100)	
q	-0.036 (-1.410)	1.350 (6.440)***		0.063 (1.850)*	0.031 (3.530)***		-0.037 (-3.020)***	0.009 (4.430)***	
BOOK	0.531 (6.770)***	0.104 (3.150)***		0.214 (2.220)**	-0.109 (-1.450)		0.320 (7.560)***	0.070 (4.240)***	
LEVERAGE									
ROA	-0.012 (-10.000)***	-0.006 (-12.850)***		-0.005 (-3.730)***	-0.004 (-3.690)***		-0.007 (-10.340)***	-0.003 (-12.290)***	
FIRM_AGE	-0.002 (-6.380)***	0.000 (0.130)		-0.002 (-4.650)***	0.002 (2.920)***		-0.001 (-6.830)***	0.000 (-0.520)	
CASH CAPITAL	0.000 (-0.060)	0.000 (-1.500)		0.000 (-0.990)	0.000 (-1.740)*		0.000 (0.280)	0.000 (-1.180)	
STOCK RETURNS	0.000 (4.120)***	0.000 (6.810)***		0.000 (-0.060)	0.001 (5.570)***		0.000 (3.110)***	0.000 (9.260)***	

Table 6 (Continued)

Dependent variable	Log(total volatility)			Log(systematic volatility)			Log(idiosyncratic volatility)		
	PCF high q [1]	PCF low q [2]		PCF high q [3]	PCF low q [4]		PCF high q [5]	PCF low q [6]	
CEO_AGE	-0.004 (-1.440)	-0.001 (-0.480)		-0.004 (-1.110)	0.001 (0.260)		-0.002 (-1.270)	0.000 (-0.200)	
$CEO_AGE*CAPEX$	0.001 (0.190)	-0.001 (-0.300)		0.011 (1.040)	0.000 (0.060)		0.000 (-0.150)	-0.001 (-0.540)	
$CEO_AGE*R\&D$	-0.034 (-1.440)	-0.036 (-2.920)***		-0.075 (-1.360)	-0.079 (-2.790)***		-0.015 (-0.960)	-0.012 (-1.960)**	
N	9938	5041		9938	5041		9938	5041	
R^2	0.605	0.482		0.655	0.397		0.564	0.490	

Table 7 Sensitivity of firm risk on CEO age and R&D by the Herfindahl Index

The sample firms are from years 1993 to 2013. All accounting variables are from Compustat. *CEO_AGE* is the age of the CEO, as reported by ExecuComp. Independent and dependent variables are defined as in Table 3. Regressions are estimated with firm fixed effect, year fixed effect, industry fixed effect (two-digit SIC code). The standard errors are robust to heteroskedasticity and clustered across firms. *t*-statistics are provided in parentheses. We calculate the median HI across all industries every year. Firms with HI above the median HI are "High HI" firms. Firms below this measure are "Low HI firms." Asterisks indicate the level of significance: *significant at 10%, **significant at 5%, ***significant at 1%.

Dependent variable	Log(total volatility)		Log(systematic volatility)		Log(idiosyncratic volatility)	
	Low HI [1]	High HI [2]	Low HI [3]	High HI [4]	Low HI [5]	High HI [6]
Constant	-5.982 (-24.690)***	-6.160 (-25.930)***	-3.853 (-12.040)***	-4.075 (-11.520)***	9.146 (72.190)***	8.968 (76.980)***
Lag residual	-0.318 (-3.260)***	0.145 (1.850)*	-0.342 (-2.580)***	0.324 (2.680)***	-0.191 (-3.600)***	-0.014 (-0.360)
VEGA						
Lag residual	-0.025 (-3.540)***	0.007 (0.740)	-0.026 (-2.610)***	0.006 (0.440)	-0.011 (-2.990)***	0.002 (0.450)
DELTA						
TENURE	0.002 (0.790)	0.005 (1.940)*	0.000 (0.140)	0.009 (2.610)***	0.001 (1.120)	0.001 (0.970)
CEO CASH	0.000 (-0.040)	0.000 (-0.340)	0.000 (0.500)	0.000 (-0.360)	0.000 (0.160)	0.000 (-0.120)
COMPENSATION						
SIZE	-0.263 (-8.580)***	-0.214 (-8.290)***	-0.131 (-3.300)***	-0.031 (-0.750)	-0.159 (-10.060)	-0.132 (-10.620)
SALES GROWTH	0.107 (4.280)***	0.062 (3.540)***	0.094 (2.690)***	0.101 (3.590)***	0.062 (4.790)***	0.033 (3.510)***
R&D	0.328 (0.090)	2.088 (1.900)*	-2.487 (-0.510)	4.118 (2.610)***	0.453 (0.290)	0.966 (1.850)*
CAPEX	-0.003 (-0.010)	0.212 (1.040)	-0.166 (-0.460)	0.340 (0.980)	0.015 (0.110)	0.057 (0.570)
q	0.010 (1.130)	0.013 (2.130)**	0.047 (3.370)***	0.026 (2.730)***	-0.003 (-0.670)	0.001 (0.190)
BOOK LEVERAGE	0.488 (6.880)***	0.124 (2.310)**	0.151 (1.610)	-0.103 (-1.260)	0.271 (7.180)***	0.096 (3.610)***
ROA	-0.013 (-9.980)***	-0.005 (-6.170)***	-0.005 (-3.540)***	-0.003 (-3.550)***	-0.008 (-10.060)***	-0.003 (-5.850)***
FIRM_AGE	-0.001 (-2.160)**	0.000 (-0.060)	0.000 (-0.130)	0.001 (1.340)	-0.001 (-2.320)**	0.000 (-0.560)
CASH CAPITAL	0.000 (0.330)	0.000 (-4.020)***	0.000 (0.120)	0.000 (-1.980)***	0.000 (-0.450)	0.000 (-3.740)***
STOCK RETURNS	0.000 (8.710)***	0.000 (-2.800)***	0.000 (2.390)**	0.000 (-16.970)***	0.000 (8.870)***	0.000 (-2.660)***
CEO_AGE	-0.002 (-0.680)	-0.002 (-0.600)	0.000 (-0.120)	-0.002 (-0.480)	-0.001 (-0.900)	0.000 (0.080)

Table 7 (Continued)

Dependent variable	Log(total volatility)		Log(systematic volatility)		Log(idiosyncratic volatility)	
	Low HI [1]	High HI [2]	Low HI [3]	High HI [4]	Low HI [5]	High HI [6]
CEO_AGE*CAPEX	0.001 (0.320)	-0.001 (-0.310)	0.008 (1.160)	0.000 (0.020)	0.000 (-0.120)	0.000 (-0.190)
CEO_AGE*R&D	-0.039 (-0.560)	-0.042 (-2.140)**	0.017 (0.170)	-0.076 (-2.640)***	-0.021 (-0.640)	-0.019 (-2.110)**
N	6994	8023	6994	8023	6994	8023
R ²	0.515	0.550	0.524	0.485	0.497	0.540

Table 8 This table provides descriptive statistics for our sample and control firms that result from the propensity score matching algorithm

The differences between sample and control firms are also reported. The difference in mean is tested by the *t*-test and the difference in median is tested by the Wilcoxon Sign Rank test. *p*-values are in parentheses. Asterisks indicate the level of significance: *significant at 10%, **significant at 5%, ***significant at 1%.

	Sample firms		Control firms		Difference (sample – control)	
	Mean	Median	Mean	Median	Mean	Median
<i>q</i> (<i>p</i> -value)	1.801	1.342	1.762	1.311	0.019 (0.591)	0.012 (0.272)
<i>FIRM_AGE</i>	40.458	29.000	40.743	30.000	−0.225 (0.678)	−1.000 (0.138)
(<i>p</i> -value)						
<i>SIZE</i>	7.053	6.917	7.048	6.936	0.005 (0.767)	0.012 (0.193)
(<i>p</i> -value)						
<i>R&D</i>	0.054	0.030	0.066	0.032	−0.013 (0.000)***	−0.004 (0.000)***
(<i>p</i> -value)						
<i>CEO_AGE</i>	61.465	60.000	50.857	52.000	10.608 (0.000)***	10.000 (0.000)***
(<i>p</i> -value)						
<i>N</i>	615	615	615	615	615	615

Table 9 The sample firms are from years 1993 to 2013.

Regression is estimated based on the difference between the sample and matching control firms. All variables are defined as in Table 2. Regression is estimated with firm fixed effect, year fixed effect, industry fixed effect (two-digit SIC code). The standard errors are robust to hetereskedasticity and clustered across firms. *t*-statistics are provided in parentheses. Asterisks indicate the level of significance: *significant at 10%, **significant at 5%, ***significant at 1%.

Dependent variables	<i>diff R&D</i> Sample minus control
Intercept	−0.050 (−1.220)
<i>diff Q</i>	0.001 (0.140)
<i>diff CASH_FLOW</i>	−0.298 (−2.640)***
<i>diff SIZE</i>	−0.013 (−2.150)**
<i>diff TANGIBILITY</i>	−0.016 (−0.016)
<i>diff BOOK LEVERAGE</i>	−0.010 (−1.220)
<i>diff DIV_RATIO</i>	0.002 (0.730)
<i>diff FIRM_AGE</i>	0.001 (0.490)
<i>diff SLACK</i>	−0.091 (−0.770)
<i>diff SALES_GROWTH</i>	0.023 (4.910)***
<i>diff BONUS</i>	0.000 (−0.030)
<i>diff CEO_EQUITY</i>	0.011 (1.320)
<i>diff TENURE</i>	−0.015 (−1.730)*
<i>diff CEO_AGE</i>	0.005 (0.840)
<i>diff CEO_AGE*q</i>	−0.019 (−2.740)***
<i>N</i>	609
<i>R</i> ²	0.1265

Table 10 The sample firms are from years 1993 to 2013. Variables are defined as in Table 3

Regressions are estimated based on the difference between the sample and matching control firms. Regressions are estimated with firm fixed effect, year fixed effect, industry fixed effect (two-digit SIC code). The standard errors are robust to heteroskedasticity and clustered across firms. *t*-statistics are provided in parentheses. Asterisks indicate the level of significance: *significant at 10%, **significant at 5%, ***significant at 1%.

Dependent variable	Diff <i>TOT_VOL</i>	Diff <i>SYSVOL</i>	Diff <i>IVOL</i>
Sample minus control			
Constant	-0.147 (-0.410)	0.060 (1.080)	-0.112 (-0.630)
Diff lag residual <i>VEGA</i>	0.269 (0.830)	-0.031 (-0.070)	0.188 (0.188)
Diff lag residual <i>DELTA</i>	0.097 (2.780)***	0.031 (0.340)	0.059 (3.010)***
<i>diff TENURE</i>	0.015 (1.290)	0.019 (1.190)	0.008 (1.370)
<i>diff CEO CASH</i>	0.000 (-2.670)***	0.000 (-2.680)***	0.000 (-2.330)***
<i>COMPENSATION</i>			
<i>diff SIZE</i>	-0.172 (-1.710)*	-0.083 (-0.750)	-0.096 (-1.830)*
<i>diff SALES GROWTH</i>	0.096 (1.120)	0.140 (1.300)	0.046 (0.046)
<i>diff R&D</i>	-2.574 (-1.840)*	-1.461 (-1.030)	-1.272 (-1.970)**
<i>diff CAPEX</i>	0.131 (0.480)	-0.181 (-0.590)	-0.087 (-1.040)
<i>diff q</i>	0.045 (1.950)*	0.062 (1.590)	0.019 (1.390)
<i>diff BOOK LEVERAGE</i>	0.245 (0.760)	-0.509 (-1.000)	0.207 (1.470)
<i>diff ROA</i>	-0.006 (-2.500)**	-0.003 (-0.860)	-0.003 (-2.370)**
<i>diff FIRM_AGE</i>	-0.003 (-1.360)	-0.001 (-0.610)	-0.001 (-1.010)
<i>diff CASH CAPITAL</i>	0.000 (-0.340)	0.000 (-0.050)	0.000 (-0.570)
<i>diff STOCK RETURNS</i>	0.001 (1.300)	0.001 (1.720)*	0.000 (1.020)
<i>diff CEO_AGE</i>	-0.008 (-0.580)	-0.019 (-0.830)	-0.004 (-0.520)
<i>diff CEO_AGE*CAPEX</i>	0.002 (0.140)	-0.028 (-1.530)	0.000 -0.330
<i>diff CEO_AGE*R&D</i>	-0.192 (-2.800)***	-0.226 (-3.090)***	-0.083 (-2.490)***
<i>N</i>	435	435	435
<i>R</i> ²	0.060	0.060	0.046

provide a predicted probability known as the propensity score. For every sample firm there may be one or more potential control firms in the same portfolio. Sample firms are matched to the closest control firms based on the propensity score. The stringent requirements of the matching process reduces substantially the number of observations, and we are left with a set of 615 matched firms.

Descriptive statistics of the matched sample may be seen in Table 8. The effectiveness of the matching procedure may be seen in the first three variables, which describe the matching variables of the subsamples. In the cases of *q*, *FIRM_AGE*, and *SIZE*, the sample and control firms display consistent similarity. The difference between sample and control firms for all these three variables are insignificant, indicating a good match. By contrast, both *R&D* and *CEO_AGE* exhibit statistically significant differences. Obviously, *CEO_AGE* is quite different by construction. *R&D* also exhibits significant differences. Sample firms (with older CEOs) exhibit both

lower mean and median R&D expenditures compared to the control firms (with younger CEOs) even though they are similar in terms of *FIRM_AGE*, *SIZE*, and *q*.

We next repeat the regressions from equation (5) based on the difference between the sample and control firms. The results are presented in Table 9. It can be seen in Table 9 that the difference in R&D between the sample and control firms is driven by the difference in the interaction term of *CEO_AGE* and *q*. Given that the sample firms are similar to the control firms in terms of various firm characteristics but different in terms of the CEO age, we can attribute the difference in R&D expenditure to the difference in the interaction term of *CEO_AGE* and *q*. Older CEOs disrupt the relationship between R&D expenditure and growth options (proxied by *q*) to a greater extent than younger managers, consistent with our earlier findings.

We also repeat the regressions from Table 3, in which we regress the independent variables on our three measures of firm risk: *SYSVOL*, *IVOL*, and *TOT_VOL*, defined in equations (2–4). We estimate these regressions based on the difference between the sample and control firms. The results are presented in Table 10. The sample and control firms are similar with respect to various firm-level characteristics but are different in terms of *CEO_AGE*. Again we find that the differences in the three measures of firm risk are driven by the difference in the interaction term between *CEO_AGE* and *R&D*. These results are consistent with the results reported in Table 3.

Our findings are robust to the above-mentioned endogeneity biases. Our results suggest that the differences in R&D and differences in firm risk are not driven by any firm-level characteristics like *FIRM_AGE* but by the interaction of *CEO_AGE* with *q* and *CEO_AGE* with *R&D* respectively.

5. Conclusion

It has long been asserted that older CEOs reduce firm risk. This has been established most recently and meticulously by Serfling (2014). We verify this assertion by establishing a link between older CEOs and risk as measured from firm returns. However, when we estimate this link in such a way as to allow for the indirect channel from CEO age to firm risk to run through R&D expenditures, the direct association between CEO age and firm risk is no longer present, and only the indirect channel remains.

Further, when we examine R&D expenditures, we find that the age of the CEO is associated with a distortion in the relationship between R&D and growth options, as measured by Tobin's *q*. Long-established economic literature demonstrates that there should exist a direct link between firm investment and growth opportunities, implying that this distortion is suboptimal for the profitability of the firm. When, for a given growth opportunity, older CEOs invest less in the risky expenditures that comprise research and development, they sacrifice firm value.

However, Serfling (2014) asserts that firms may hire older CEOs specifically *because* they are expected to reduce firm risk. If this is the case, then we would expect

transparent, well-governed firms to exhibit a direct or indirect relationship between CEO age and firm risk. However, when we control for a range of corporate governance measures we find that well-governed firms do not exhibit a relationship between CEO age and firm risk. We find that poorly governed firms are driving the overall findings, and that these are the firms that exhibit associations between CEO age and firm risk—and that these associations are primarily through the channel of R&D expenditures. Further, we find that firms with more effective governance (as measured by a range of governance measures including earnings management, the coincidence of free cash flow and Tobin's q , and a Herfindahl Index of sales) do not exhibit a distortion from CEO age in the association between growth opportunities and R&D expenditures. Nor do they exhibit any direct association between CEO age and R&D expenditures. These findings suggest that the effects of CEO age on firm risk and on R&D expenditures are not beneficial to the firm.

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